

Assembly

Assembling Link 1A and Link 1B

Assembling Link 2A and Link 2B

M2.5-5 screws (inc. with XM540) are required and it appears not all screws in the bag are of this exact length (or there is a tolerance issue with my Frame A). Make sure to use the correct length screws to secure Frame A to the servos (inner radius holes are used, which are unthreaded, and threads engage below on the servo horn).

There is an internal duct on one side through which the cable must be passed. This should be done before putting sides 2A and 2B together. It is definitely possible to push the connector through but not easy. It will usually get stuck but can be done without any tooling.

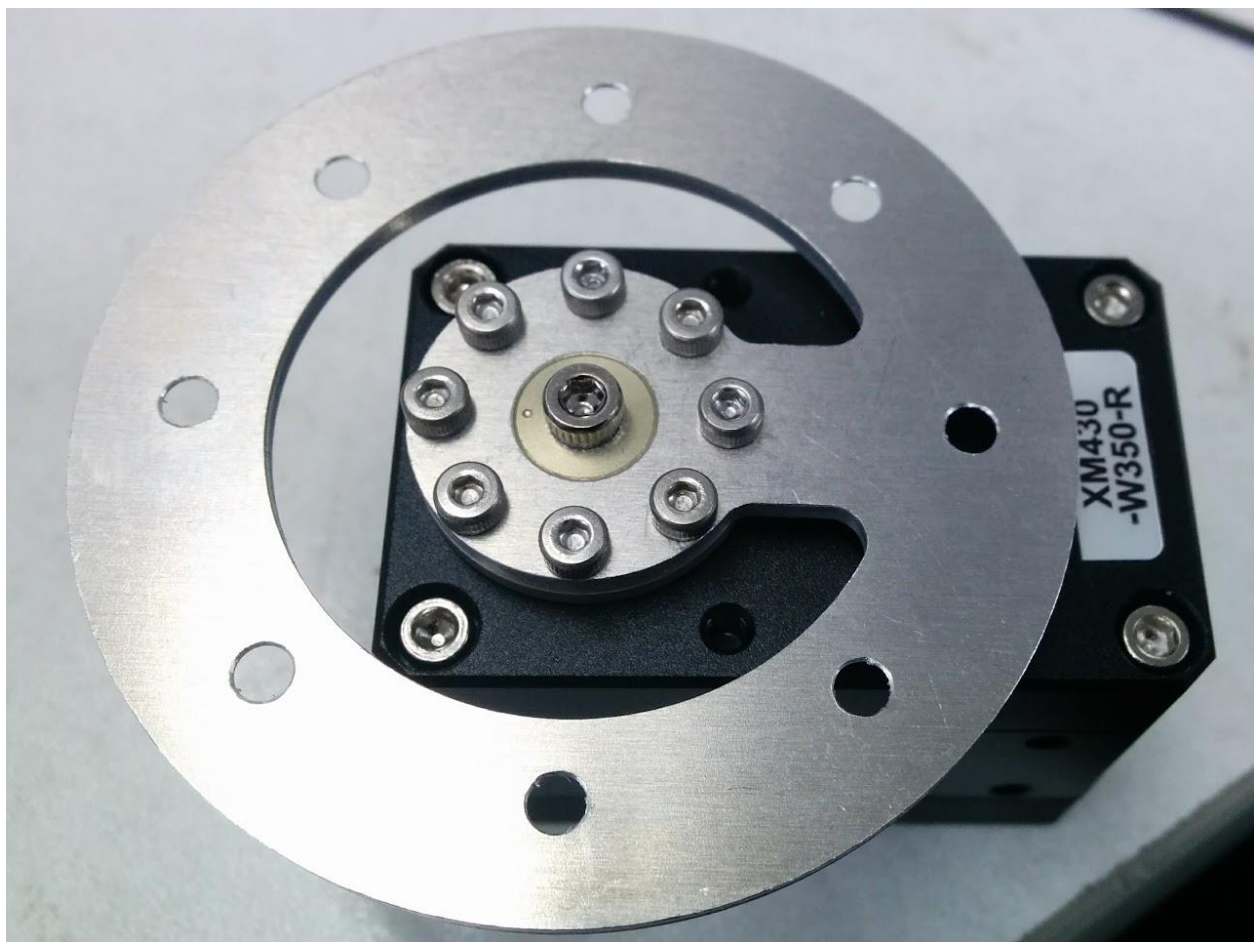


M3x6 screws (UF-0306N) were used to secure the link to Frame A on each side.



Assembling the Base

To affix Frame C onto the servo horn (which must be attached with spacer and the screw included with the motor), 8 M2x4 screws are needed (to go through Frame C, as shown below, and into the threads in the servo horn).

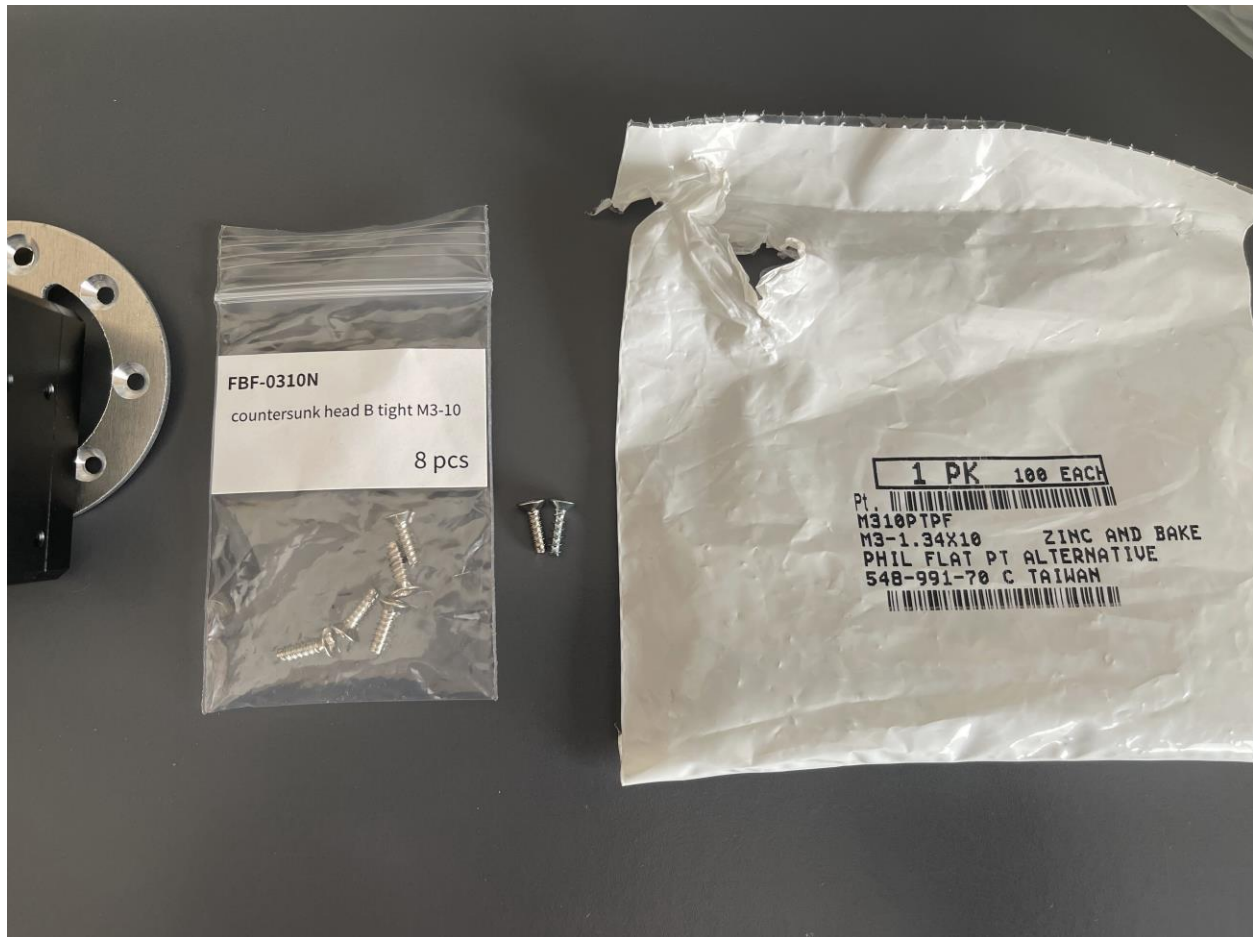


The screws included with the XM430-W350-R motor appear to be too short, M2x3? I found the following M2x4 screws on Amazon with hex key socket head knurl cap:

https://www.amazon.com/dp/B011BNU6TY?psc=1&ref=ppx_yo2ov_dt_b_product_details



For attaching Frame C to the plastic of Link 1, M3x10 self-tapping screws with a flat 90 degree countersunk head are required. The Japanese part, FBF-0310N is almost equivalent to M3x10 self-tapping screws with a 1.34mm thread pitch from McMaster-Carr (part number [90434A129](#)). The latter have a coarser thread pitch than the Japanese screws.



Japanese parts on the left (FBF-0310N), US sourced equivalent on the right (90434A129).

Attach the bearing to the bottom of Link 1. Remember to reset servo position before screwing Frame C to the plastic of Link 1, as below. The position of the servo will have rotated while affixing Frame C to the horn and securing it with the 8 M2x4 screws.



Note the cable from motor ID is passed through the inside radius of Frame C.

The base is assembled over motor ID 2. Make sure to properly mount the PLT connector and attach the other end of the cable to the motor. Make sure the washer and nut are securely screwed in on the inside.



Finally, tighten the two sides of the base using M3x10 tapping screws with a pan head (FBP-0314N). Should be possible to find US-sourced parts but they should not be countersunk and have a flat underside, like the pan head screws in the image below.



Assembling Link 3



6809ZZ attached to link 2. Use M2x4 screws as before to secure Frame D to motor ID4's servo horn and then M2.6-10 countersunk screws (FBF-2610N) to attach the frame with motor to Link 2.



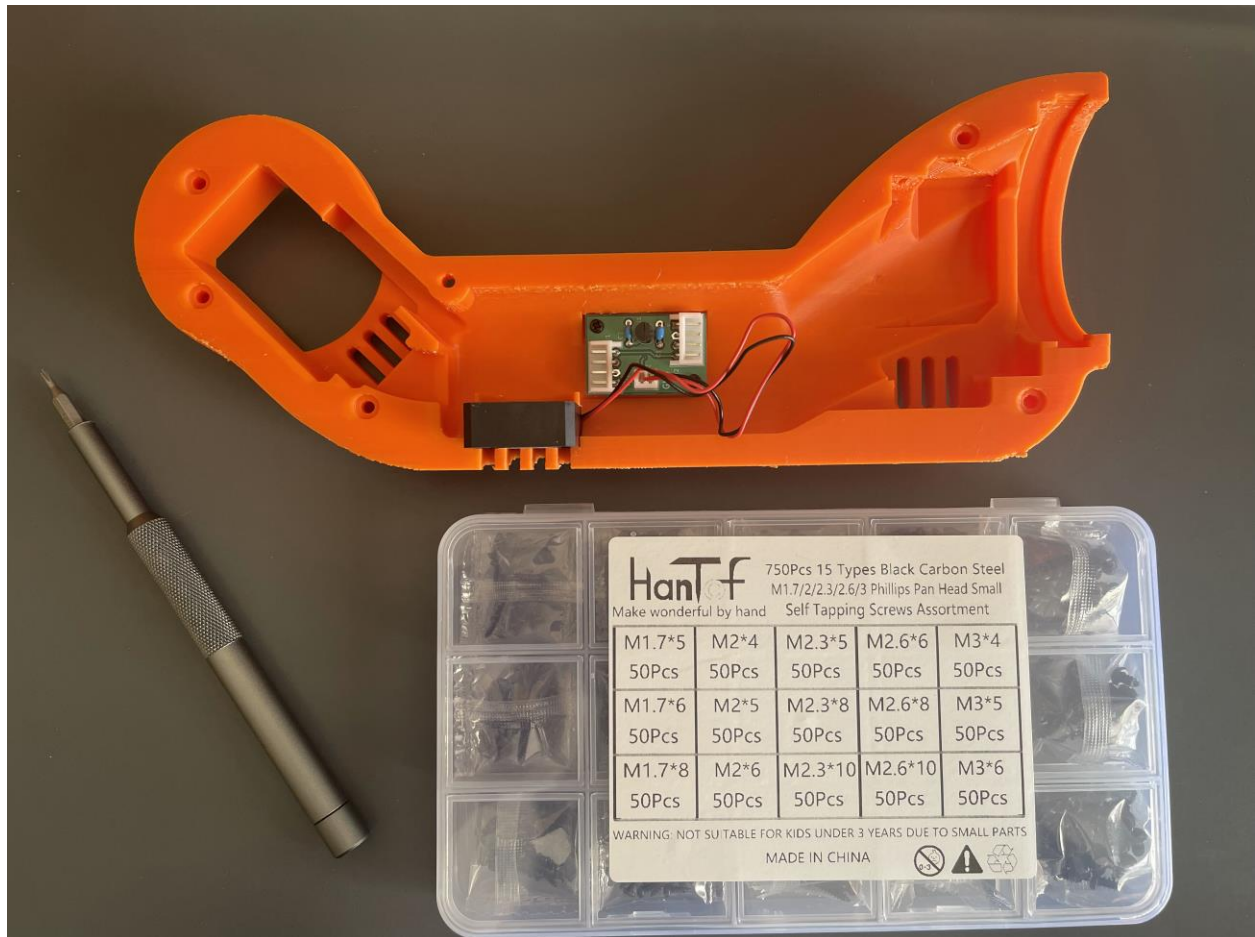
Next, the fan board must be mounted inside of Link 3.

Fan board:

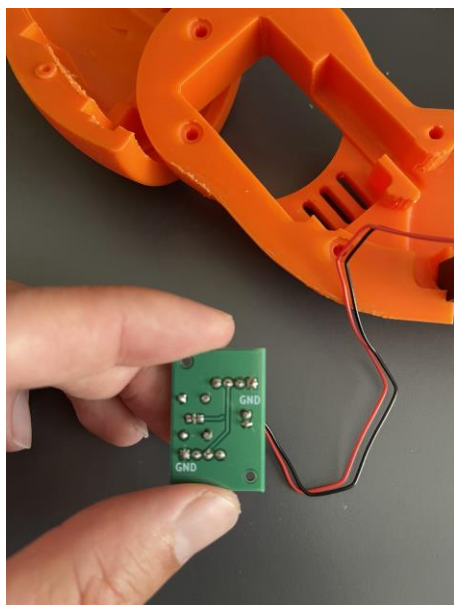
- Pass through of 4-pin motor connectors.
- Has a 78L05 voltage regulator that supplies the fan Vdd/Gnd pins.
- C1 is marked 334.
- C2 is slightly smaller, no markings.

Use M2x5 tapping screws (pictured is this item from Amazon:

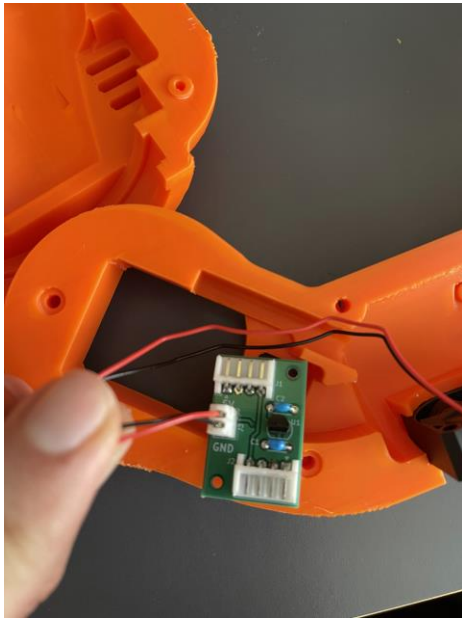
https://www.amazon.com/dp/B07ZH9GJWP?psc=1&ref=ppx_yo2ov_dt_b_product_details)



Underside of fan board:



Top side of fan board:



Motor ID 5 must have HN12-I101 idler mounted on the back. This requires taking apart the back side of the motor to pass one of the cables through the idler's axial hole.



The cable should be passed through as such (note that a small tab must be removed with pliers in order to be able to close the case again).



Finally, mount into Link 3. I used electrical tape for cable management. Small zip ties would probably be better.

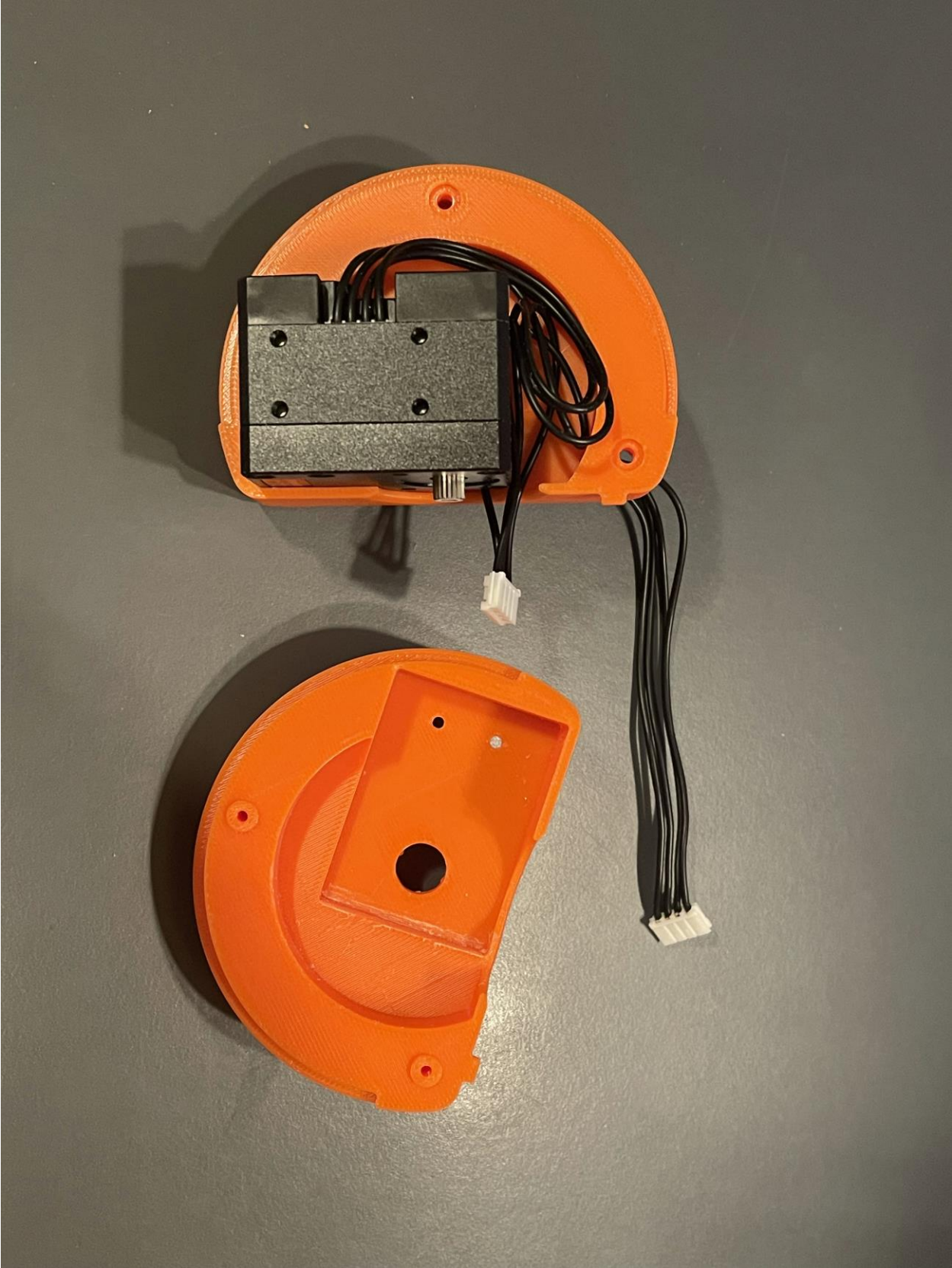


Fitting the two sides of the linkage together may require some force but must be done carefully. File down surfaces to help the motors fit.



Mount frame B onto the servo horn of motor ID5. Ensure that it fits flat and flush against the servo horn (the motor must be right up against the side wall of the linkage to be lined up with the opening correctly).

Assembling Link 6



Inserting motor ID8 into Link 6 will be an extremely tight fit! I had to shave down top insert edges.



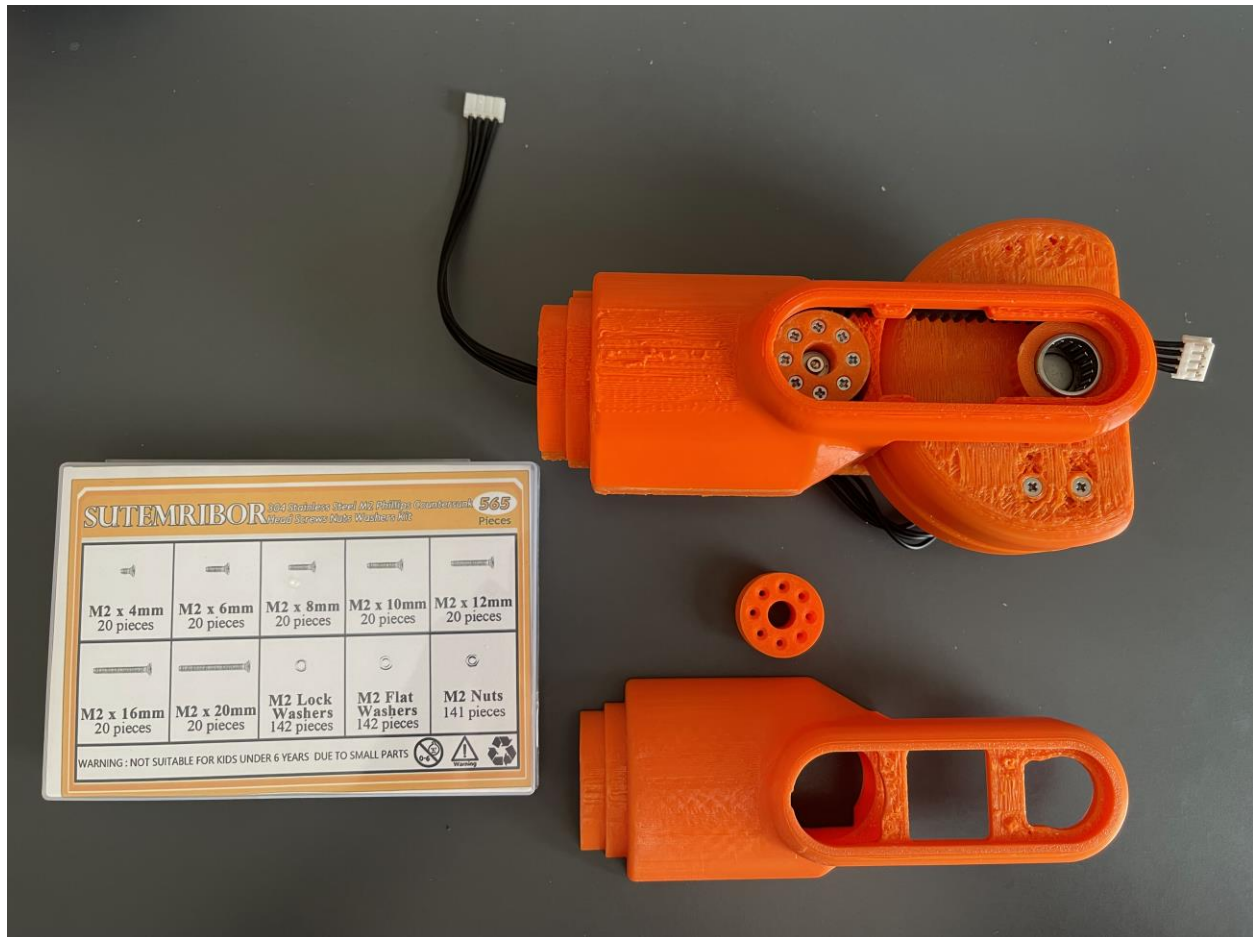


Assembling Link 5

Attachment of Link 5B and HK1612 bearing:



Version 1.2 Link 5A part is very difficult to work with. I used Version 1.0's Link 5A part which corresponds to the assembly instructions on RT Robotics' web site. M2x10 screws with a countersunk head (from here: https://www.amazon.com/dp/B0CLMBB5TM?psc=1&ref=ppx_yo2ov_dt_b_product_details) were used to secure the sprocket (Link 5E) to the servo horn. In the image below, Link 5A v1.0 is printed with PETG (I ran out of ABS). 5E was printed with both PETG and ABS but the ABS version was found to allow the screws to penetrate deeper. M2x11 would probably be a more optimal length. It is possible (but untested) that M2x12 coupled with the PETG sprocket would work better.



To mount the belt and sprocket with Link 5A, roughly the following procedure was used:

- Insert belt into Link 5A
- Holding Link 5A upright, place sprocket (Link 5E) upright onto the belt in roughly the position it will end up in
- Ensure the sprocket engages with the belt teeth
- Position the sprocket so that the screws will line up perfectly with the servo horn's screw holes (one screw should be exactly at the 3:00 position, as in the image above)
- Carefully holding the sprocket (to prevent it from falling out the bottom), start to place Link 5A over Link 5B
- Push the timing belt so that it hooks around the Link 6 joint. This will move the sprocket to the right (relative to image above), so that it is no longer aligned with servo horn.
- Using a screw driver inserted into the axial hole, pull the sprocket back and set it into place.

Inserting the torsion spring:

1. Hook the left arm under and around the belt. The spring itself should be under the belt. This step is easily accomplished and it should be easy to pull back the belt enough to do so.

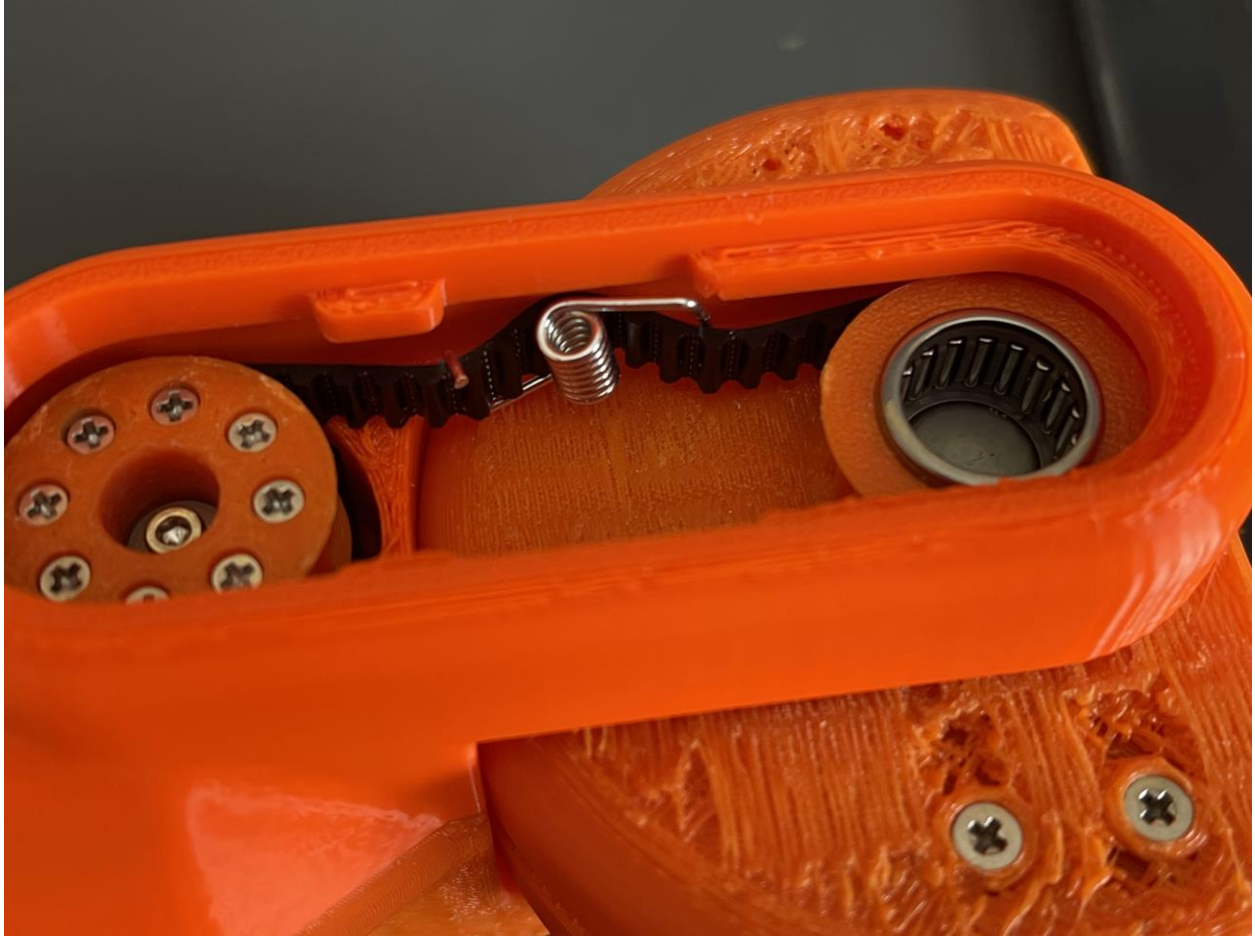
2. Next, insert something between the outside of the belt and Link 5A temporarily, to free up your hands.
3. Using pliers, grasp the right arm and twist it out (Z axis in the image) and over the belt, just enough to hook it over the belt.
4. Take the pliers and grip the belt and pull upwards while using your free hand to push down on the right arm so that it properly hooks around and sets in place. Before doing so, you may need to slide the spring left or right so that it is properly centered (and the sprocket can turn +90 and -90 degrees).



Left arm of torsion spring in place, hooked around over the belt.



Screwdriver inserted between Link 5A frame and belt to allow right arm to be pulled around the top of the belt.



Right arm positioned over belt, but not yet fully engaged (the last 90 degree bent segment is pushing into the center of the belt).



Pulling up on the belt with pliers. The right arm of the spring can then be pushed down so that it embraces the timing belt.



Final configuration. Note the positioning of the torsion spring. The servo motor home position is pointing exactly to the right and Link 6 must be straight. In this position, it is possible to rotate +90 and -90 degrees without the spring interfering with either joint.

Connecting Link 3 and Link 4 (Hand Side)

Attach Frame E to motor ID6 using M2x4 screws. Do not use the motor's included screws, which are too short, but rather the screws from Amazon with hex key socket head knurl cap:

https://www.amazon.com/dp/B011BNU6TY?psc=1&ref=ppx_yo2ov_dt_b_product_details

See below.



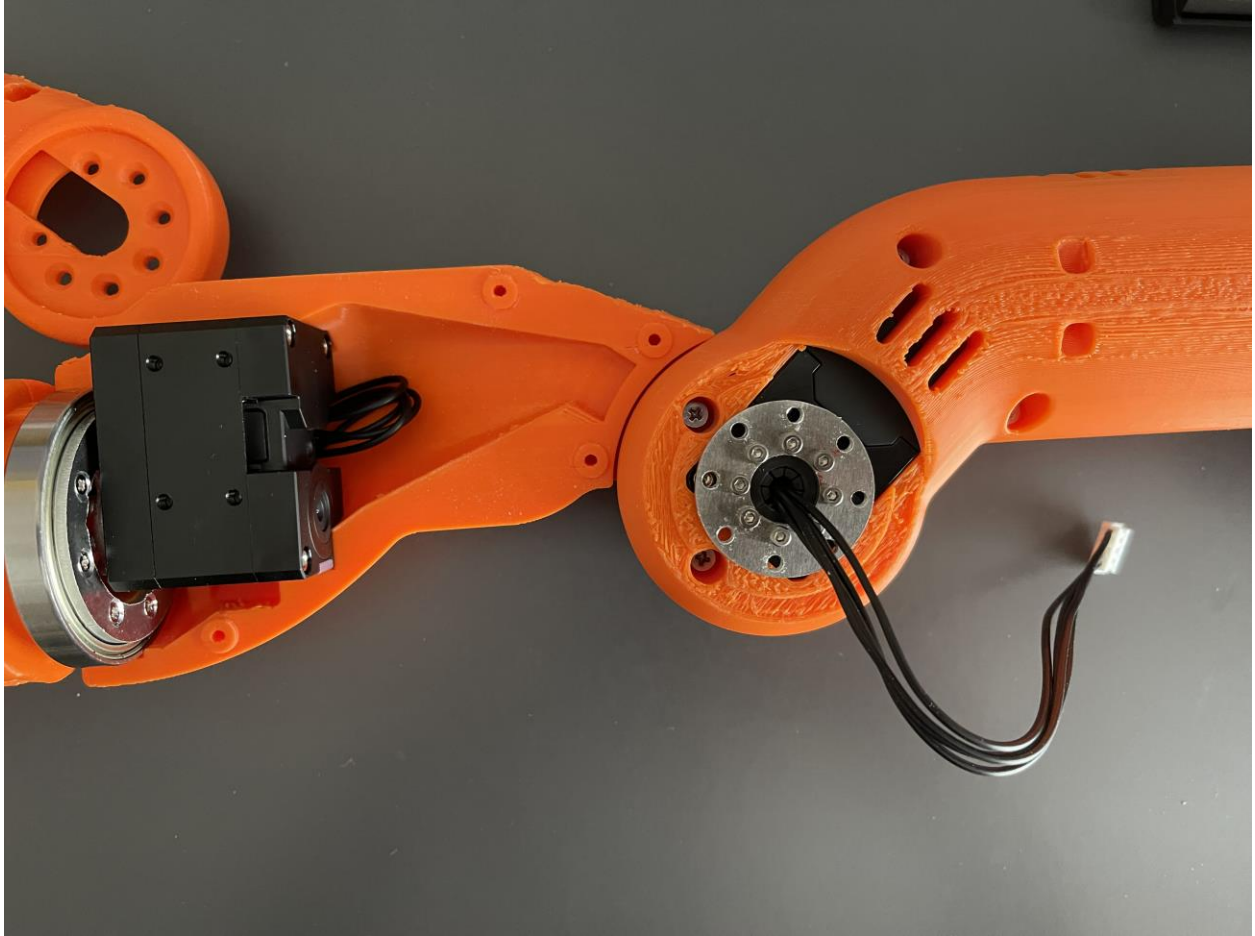
Mount 6808ZZ bearing onto link 5 and then attach frame E to link 5 with countersunk M2x10 tapping screws (FBF-0210N in this case).



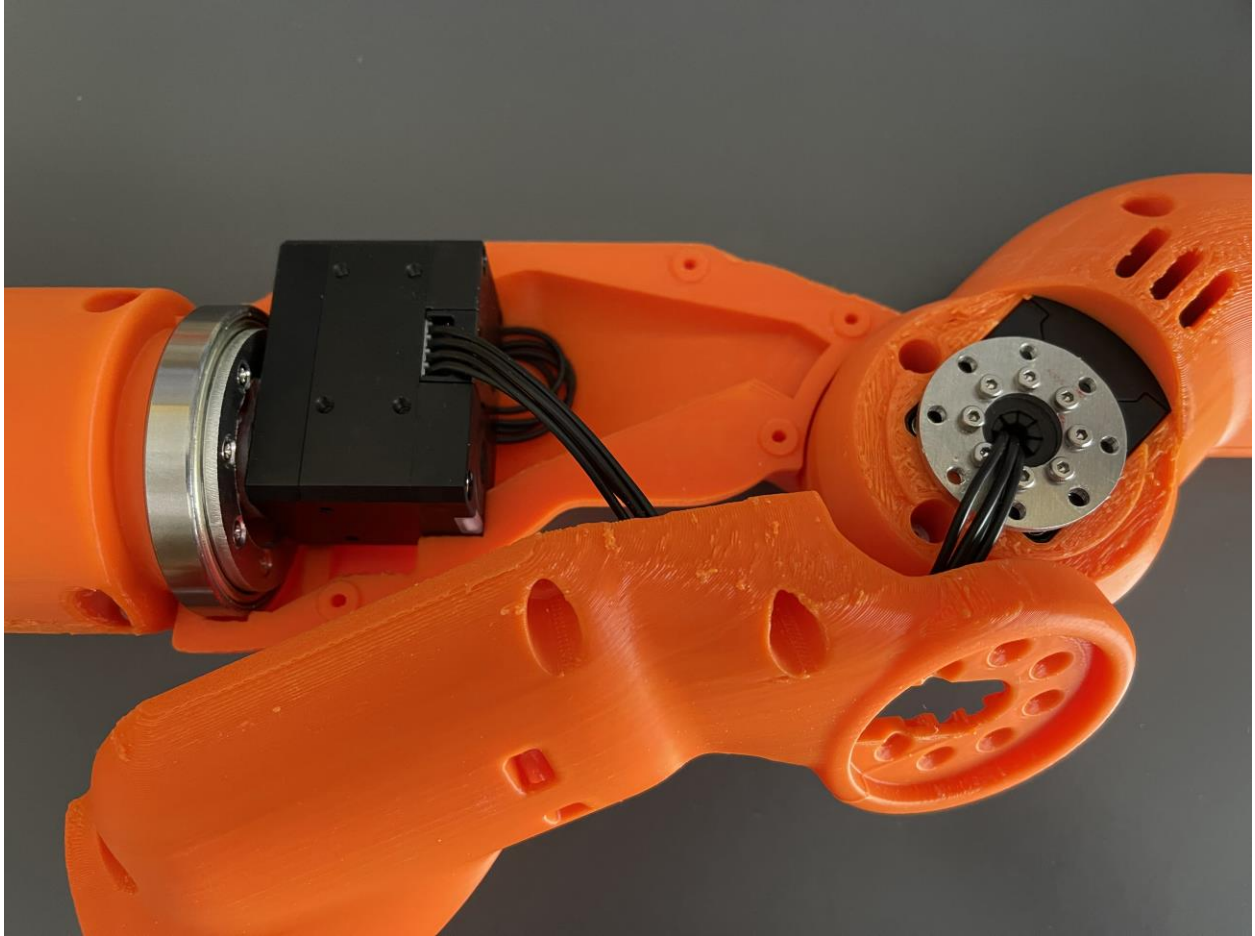
Connect link 3B and 4B in the orientation shown below using 8 M3x6 countersunk screws. Such screws can be found on Amazon but beware of hex socket screws, which are hard to remove. We use the Japanese UF-0306 screws here.



Next, place motor ID6 into Link 4A.



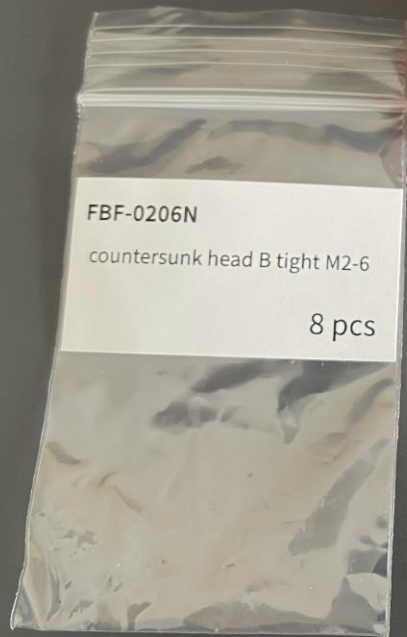
Pass the cable from the Link 3 end through Link 4B (there is a duct for it).



Use the M2x6 screws to connect the other side of the link to the servo and M3x10 tapping screws to secure the two sides of Link 4 to each other.



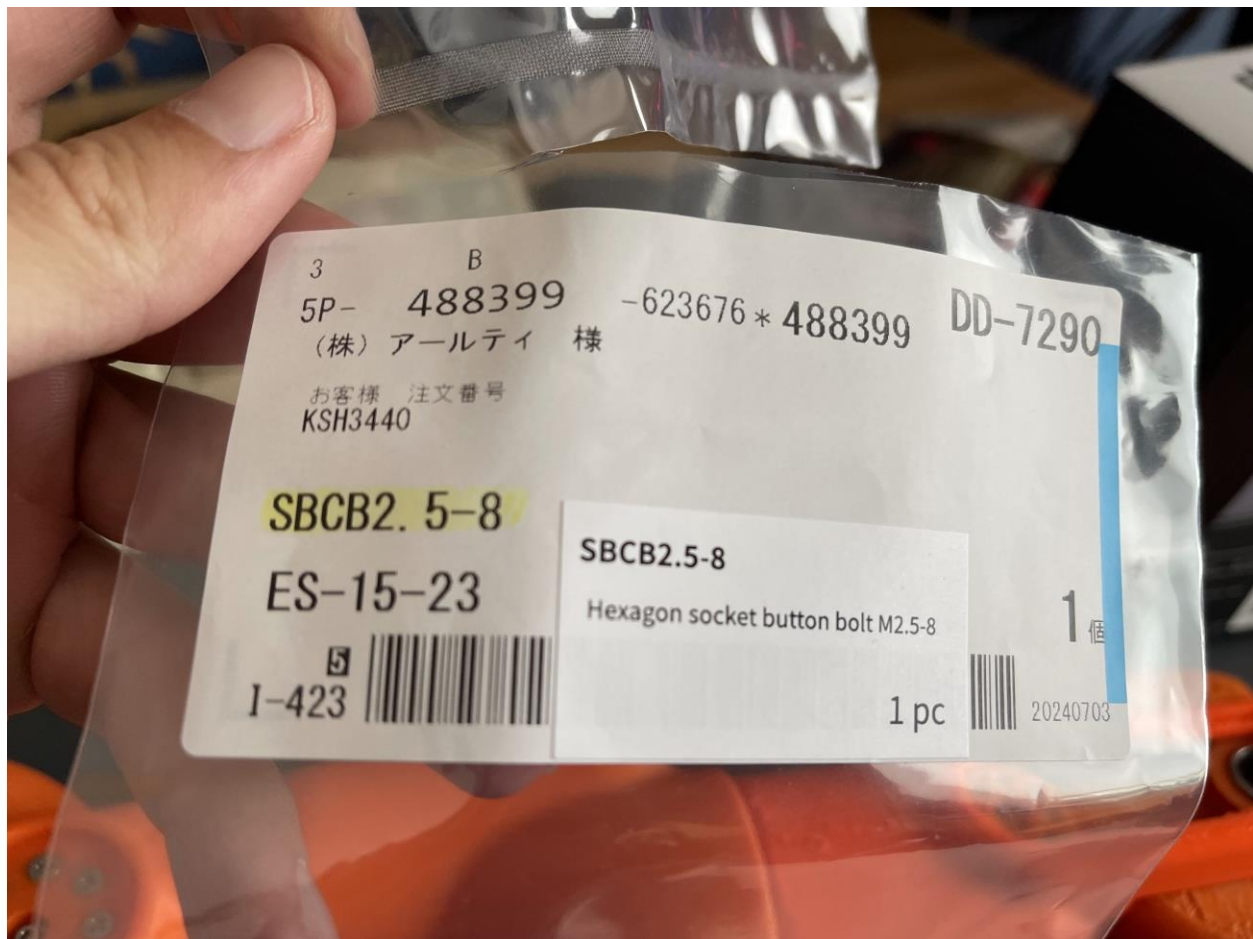
Use countersunk M2x6 screws (FBF-0206N – tapping?) to connect the hand frame to Hand A.





Original instructions call for replacing the servo horn screw with M2.5x8 hexagon socket button bolt (not sure why), as below. But **this is not recommended as this screw is difficult to remove.**





Instead, secure the servo to the frame using M2x5 screws. **Do not use the CSPSL-SUS-M2-5 (cross-recessed (+) slim head M2-5) screws!** These are very difficult to remove and strip easily. Instead, use M2x5 hex bolt cap head screws as there is enough clearance for them (Amazon link: https://www.amazon.com/dp/B07PZ7X3ND?ref=ppx_yo2ov_dt_b_product_details&th=1).



How ID9 servo should be connected and then mounted (with cables passed through the inside of the hand frame):



Picture with replacement machined frame:







Attach Hand B.



Pan-head M2x10 tapping screws secure hand A and hand B together. They are inserted into the two holes downward as shown below. Alternatively, may want to use hex bolt cap head screws (M2x10) for easier removal

(https://www.amazon.com/dp/B00YBMROKC?psc=1&ref=ppx_yo2ov_dt_b_product_details).



To attach Hand D1, the Japanese instructions refer to M2x5 flathead screws, which were not available. Instead, countersunk M2x6 screws work

(https://www.amazon.com/dp/B0CLMBB5TM?psc=1&ref=ppx_yo2ov_dt_b_product_details).





FBF-0206N
Part B



Hand D3 mounts into this exposed hole and is secured to Hand D1 with pan-head M2x10 tapping screws.



Lastly, mount Hand2 and Hand4 together and use an M3x8 tapping screw to secure (cross-recessed panhead tapping screw).

